

Immunotherapy

Immunotherapy is treatment that uses the body's immune system, usually your own, to attack cancer. Learning about how it works and what to expect can often help you prepare for treatment and make informed decisions about your care.

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What is immunotherapy?

Immunotherapy is a type of cancer treatment that helps the immune system find and attack cancer cells. It uses substances made by the body or in a lab to:

- Stimulate or boost the immune system so it works better against cancer.
- Add lab-made substances that act like natural parts of the immune system.

There are many different types of immunotherapy, and they work in different ways. The type used depends on the cancer. Immunotherapy is often used for advanced cancers, especially when other treatments haven't worked.

[Watch this video on YouTube¹](#)

How the immune system works against cancer

Your immune system is made up of organs, cells, and substances that help protect you from infections and some diseases by finding and destroying germs like bacteria and viruses. It keeps track of what normally belongs in the body. When it finds something that it either doesn't recognize, or recognizes as "foreign," the immune system raises an alarm and attacks it. This is called the **immune response**.

The immune system can also respond to cancer if it sees the cancer cells as foreign. This works best when there are only a few cancer cells. But there are limits on the immune system's ability to destroy cancer on its own:

- The immune system might not see cancer cells as foreign if they are too similar to normal cells.
- The immune response might not be strong enough to destroy the cancer.
- Cancer cells can give off signals that help them hide from the immune system.

Immunotherapy helps overcome these challenges by helping your immune system recognize cancer cells and strengthening your immune response.

Types of cancer immunotherapy

Several types of immunotherapy used to treat cancer, and many are being studied.

- **Monoclonal antibodies (mAbs or MoAbs)** are lab-made proteins designed to target specific parts of cancer cells.
- **Chimeric antigen receptor (CAR) T-cell therapy** modifies the T cells in your blood to make them better able to target cancer cells.
- **Tumor infiltrating lymphocytes (TILs)** are special white blood cells taken from a tumor, multiplied in a lab, and returned to the body to target and destroy the cancer.

- **Immune checkpoint inhibitors** remove the immune system's "brakes, allowing it to recognize and attack cancer.
- **Cancer vaccines** are used to start an immune response against certain types of cancer.
- **Cytokines** are small proteins that send signals to immune cells to attack cancer.
- **Oncolytic virus therapy** uses lab-modified viruses to infect and kill certain tumor cells.
- **Immunomodulators (non-specific immunotherapies)** boost parts of the immune system to treat certain types of cancer.

What is the goal of immunotherapy?

If your doctor has recommended immunotherapy, it's important to understand the goal of treatment. This can depend on your type of cancer, its stage (size and location), and how far it has spread. Immunotherapy can have 3 possible treatment goals:

To cure the cancer (curative intent)

Whenever possible, the goal is to cure the cancer by destroying it completely so it doesn't come back. While immunotherapy can result in long-lasting treatment responses, it doesn't work for every cancer or every person. Because it's a newer approach, doctors are still learning about its long-term outcomes.

To control the cancer

If a cure isn't possible, immunotherapy might be used to shrink the cancer or slow its growth. This can help people feel better and live longer. The [cancer might not go away completely²](#), but it can often be managed like a chronic condition such as heart disease or diabetes.

To ease symptoms (palliation or palliative intent)

Immunotherapy might be given to improve comfort and quality of life, even if the cancer has spread or can't be controlled. For example, it might help shrink a tumor that's causing pressure or pain. This is called **palliative immunotherapy**. It might be just one part of [palliative care](#)³, which includes things like pain relief and anti-nausea medicine. Palliative treatments can be used at any stage of cancer.

How is immunotherapy different from chemotherapy?

[Chemotherapy](#)⁴ and immunotherapy are both treatments for cancer, but they are different types of drugs. Immunotherapy works with your body's immune system to kill cancer cells or limit their growth, while traditional chemo works against cancer cells (and some healthy cells) directly.

Like chemo, immunotherapy is most often a **systemic treatment**. It travels through the bloodstream and can help your immune system target and kill cancer cells that have spread (metastasized) throughout the body.

Which cancers can be treated with immunotherapy?

Immunotherapies are not yet available for all cancers. While all cancers start because of [gene changes](#)⁵ (mutation), the specific mutations or protein changes can vary. These are called biomarkers. Even people with the same type of cancer (like melanoma or lung cancer) can have different biomarkers that affect which treatments might work best.

Most often, immunotherapy targets cancers with certain biomarkers. Some immunotherapy, like tumor infiltrating lymphocyte (TIL) therapy, targets many different changes in cancer cells rather than a single change. Common gene or protein changes that immunotherapy can be used for include:

- PD-1 (programmed cell death-1) protein
- PD-L1 (programmed death-ligand 1) protein
- CTLA-4 (cytotoxic T-lymphocyte-associated protein 4)
- TMB (tumor mutational burden)
- MSI (microsatellite instability)
- CD19 (cluster of differentiation 19) protein
- CD20 (cluster of differentiation 20) protein

- BCMA (B cell maturation antigen)
- GPRC5D (G protein-coupled receptor family C group 5 member D)
- CD38 (cluster of differentiation 38) protein
- SLAMF7 (signaling lymphocyte activation molecule family member 7) protein

[Biomarker testing](#)⁶ is done before treatment to check for these or other changes to help show if immunotherapy may be an option. It may also be done if treatment fails or the cancer grows, to see if new changes have occurred.

Learn about immunotherapy for your cancer type

Choose a cancer type Acute Lymphocytic Leukemia (ALL) Anal Cancer Basal and Squamous Cell Skin Cancer Bile Duct Cancer Bladder Cancer Bone Cancer Breast Cancer Cervical Cancer Colon and Rectal Cancer Endometrial Cancer Esophageal Cancer Eye Cancer Gallbladder Cancer Hodgkin Lymphoma Kaposi Sarcoma Kidney Cancer Laryngeal and Hypopharyngeal Cancer Leukemia in Children Liver Cancer Melanoma Merkel Cell Skin Cancer Mesothelioma Nasal Cavity and Paranasal Sinus Cancer Nasopharyngeal Cancer Neuroblastoma Non-Hodgkin Lymphoma Non-small Cell Lung Cancer Oral Cavity and Oropharyngeal Cancer Ovarian Cancer Pancreatic Cancer Prostate Cancer Salivary Gland Cancer Small Cell Lung Cancer Small Cell Lung Cancer Soft Tissue Sarcoma Stomach Cancer Uterine Sarcoma Vaginal Cancer Waldenstrom Macroglobulinemia

[Go](#)

Choosing the right immunotherapy

Choosing the right immunotherapy, dose, and schedule is very important. Immunotherapy is sometimes called precision or personalized medicine because even people with the same type of cancer can have different biomarkers. Different immunotherapies are used depending on which biomarkers are found through testing. Sometimes, the treatment plan is clearly known and based on research in similar people. Other times, less is known, especially for certain cancer types or stages.

Your cancer care team will consider several factors when recommending treatment options including:

- Type and subtype of cancer
- Stage of cancer (location, size, and whether it has spread)
- Biomarker or other test results
- Age
- Overall health and current medications

- Other serious health problems (such as heart, liver, or kidney diseases) and conditions that impair your immune system (such as autoimmune disease)
- Treatment goals
- Past cancer treatments

If your treatment plan isn't clear or you are given several options, ask about [getting a second opinion](#)⁷. It's important to understand your choices and feel confident in your decision.

How does immunotherapy work with other treatments?

Immunotherapy works better for some cancers than for others. For certain cancers, it's used on its own. For others, it works better when combined with treatments such as chemo, surgery, or radiation therapy.

Immunotherapy may be used:

- **Before surgery or radiation** to shrink a tumor (**neoadjuvant therapy**)
- **After surgery or radiation** to help kill any remaining cancer cells (**adjuvant therapy**)
- **Alongside other drugs** that act directly on cancer cells, such as chemo.
- **If cancer comes back or doesn't fully respond** to other treatments.

Sometimes, different types of immunotherapy are used together. They may be given in a certain order or in certain combinations (called **combination immunotherapy**). Using more than one type can help kill more cancer cells lower the risk of the cancer becoming resistant to one type of immunotherapy.

What is it like to get immunotherapy?

Most people get immunotherapy as infusions at a clinic or hospital. Some types can be given as injections, taken by mouth, or applied to the skin. Each method has its own schedule, side effects, and safety steps.

Immunotherapy cycles

Immunotherapy can be given daily, weekly, or monthly, often at regular intervals called **cycles**. A cycle includes one or more doses followed by a break. This allows your body to respond to treatment and recover.

Each immunotherapy follows a schedule designed to maximize its effect and minimize side effects. If you're also getting chemo or more than one type of immunotherapy, your treatment plan will say how often and when each treatment is given.

The number of cycles given may be decided before treatment starts, based on the type and stage of cancer. In some cases, the number is flexible and will take into account how the treatment affects the cancer and your overall health. Ask your cancer care team how many cycles are planned and how long they expect your treatment to last.

Intravenous (IV) immunotherapy

Most types of immunotherapy are given as an infusion through an IV. The process is much like [getting chemo infusions](#)⁸. If you are getting both chemo and immunotherapy, they might be given at the same time or during separate appointments, depending on your treatment plan.

Intravesical therapy

[Intravesical therapy](#)⁹ may be used for some types of early stage bladder cancer. The immunotherapy is infused directly into the bladder using a urinary catheter.

Immunotherapy injections

Some types of immunotherapy are given as an injection or shot rather than through an IV. These shots go into specific layers of tissue:

- **Subcutaneous (SC or SQ) immunotherapy** is injected into the fatty tissue just under the skin, similar to [subcutaneous chemo](#)¹⁰.
- **Intralesional immunotherapy** is injected directly into the tumor.

Not all types of immunotherapy are able to be given as injections. Giving immunotherapy as an injection is just as effective as an IV infusion but takes less time and doesn't require IV access.

Oral immunotherapy

Oral immunotherapy is taken by mouth as a pill. Examples include lenalidomide and pomalidomide. It may be given in rounds or cycles and can sometimes be given along with traditional chemo. Because oral immunotherapy is taken at home, it's important to understand exactly how and when to take it, and what to do if you have any problems. Our information on [oral chemotherapy](#)¹¹ can help you know what to expect and what to ask your care team.

Topical immunotherapy

Topical immunotherapy is rubbed onto the skin. It is often used as a [non-surgical treatment](#)¹² for certain types of early skin cancers, such as basal or squamous cell, or precancerous changes, such as actinic keratoses. It can be given alone or in combination with other cancer treatments, like chemo. Our information on [topical chemotherapy](#)¹³ can help you know what to expect and what to ask your care team.

What are the most common side effects of immunotherapy?

Immunotherapy can cause different side effects than those from other cancer treatments because of how immunotherapy uses the body's immune system to destroy cancer.

The side effects may be different for each person. They depend on the type and dose of immunotherapy you receive, as well as your type of cancer, location, and your general health.

Most side effects are mild, however some can be severe. Some general side effects of immunotherapy include:

- [Skin changes](#)¹⁴ (such as rash)
- [Fatigue](#)¹⁵

- [Nausea, vomiting](#)¹⁶, or [diarrhea](#)¹⁷
- Endocrine changes (such as thyroid changes)
- Flu-like symptoms (such as [fever](#)¹⁸, chills, and muscle aches)
- Cytokine release syndrome (CRS)
- Neurologic toxicity
- Colitis

Immunotherapy may also cause the immune system to attack healthy cells. This can cause **immune-related adverse effects**. These may happen at any time during treatment or sometimes even after stopping immunotherapy.

It's important to be aware of possible side effects from immunotherapy so you know what to watch for and can talk with your cancer care team about preventing and relieving them.

Immunotherapy safety

Because immunotherapy is a newer type of cancer treatment, there isn't as much information about its long-term effects of exposure. To be safe, many experts recommend treating it as hazardous and taking the same [precautions used when handling chemo](#)¹⁹ or other hazardous drugs .

How successful is immunotherapy?

Immunotherapy can be successful at stimulating the immune system to kill cancer cells. However, the success rates are often lower as compared with other types of cancer treatment. This may be because:

- **Immunotherapy is often used in advanced stages of cancer** where other treatments have been unsuccessful.
- **The cancer cells can sometimes outsmart the immune system** by making substances to turn off or help them hide from the body's immune response.
- **Drug resistance can cause immunotherapy to stop working.** These can include genetic changes in cancer cells that cause the immune system to not be able to recognize or attack them, or changes in the internal and external cell environment that interfere with immune function.

- **People can respond differently to immunotherapy.** Any variation from person to person affects how well the immunotherapy works against their cancer.

When treatment plans change

It's important, whenever possible, to get the full course of immunotherapy, the full dose, and keep the cycles on schedule. This gives you the best chance of getting the maximum benefit from treatment.

There may be times, though, when serious side effects require adjusting the plan (dose and/or schedule) to allow your body more time to recover. Sometimes, you might be given supportive medicines to help you recover more quickly. Or sometimes, cancer cells develop resistance to a certain immunotherapy and a different treatment might be tried.

Still, there may be other times when treatment no longer works and there are no good treatment options available to control the cancer. Learn more about things to consider [if cancer treatments stop working](#)²⁰.

Hyperlinks

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Monoclonal Antibodies and Their Side Effects

One way the immune system protects the body is by making **antibodies**—proteins that stick to specific **antigens on germs and other foreign substances**. Antibodies circulate throughout the body until they find and attach to their target, helping the immune system destroy harmful cells. Monoclonal antibodies (mAbs) are a type of cancer treatment designed to mimic this natural process.

[What are monoclonal antibodies \(mAbs\)?](#)

[How are monoclonal antibodies used?](#)

[What are monoclonal antibodies made of?](#)

[Types of mAbs used to treat cancer](#)

[Side effects of monoclonal antibodies](#)

What are monoclonal antibodies (mAbs)?

Monoclonal antibodies (mAbs or Moabs) are lab-made antibodies (a type of immune protein) designed to that specifically target a certain antigen, such as one found on cancer cells. They are made from a single clone of a B cell, a type of white blood cell and part of the immune system. This makes them different from polyclonal antibodies that are developed from many different clones and can target multiple antigens.

How are monoclonal antibodies used?

Monoclonal antibodies are used to treat many diseases, including some types of cancer. To make a monoclonal antibody, researchers first have to identify the right antigen to attack. Finding the right antigens for cancer cells isn't always easy, and so far, mAbs have proven to be more useful against some cancers than others.

Are monoclonal antibodies immunotherapy or targeted therapy?

Different monoclonal antibodies can act as targeted therapy, immunotherapy, or both, depending on how they work.

- Some monoclonal antibodies are considered [targeted therapy](#)¹ because they attach to a specific target on a cancer cell and stop it from functioning.
- Others act as immunotherapy because they help the immune system find and attack cancer cells more effectively.

Other uses for monoclonal antibodies

In addition to treating some types of cancer, monoclonal antibodies can be used to treat other diseases and conditions, such as:

- Asthma
- Autoimmune disorders
- High cholesterol
- Infection
- Inflammatory bowel disease
- Myasthenia gravis
- Organ rejection after transplantation
- Osteoporosis

Monoclonal antibodies are also used to help diagnose certain diseases or conditions. In cancer, they're used as part of immunohistochemistry testing that looks for patterns in cancer cells. They can also be used to detect certain infections and pregnancy.

What are monoclonal antibodies made of?

Monoclonal antibodies are lab-made proteins. There are 4 different ways they can be made, and they are named based on what they are made of.

- **Murine:** Made from mouse proteins, with drug names in **-omab**
- **Chimeric:** A combination of mouse and human proteins, with drug names ending in **-ximab**
- **Humanized:** Made from small parts of mouse proteins attached to human proteins, with drug names ending in **-zumab**

- **Human:** Fully human proteins, with drug names ending in **-umab**

Types of mAbs used to treat cancer

Naked monoclonal antibodies

Naked mAbs are antibodies that have no drug or radioactive material attached to them. They work by themselves. These are the most common type of mAbs used to treat cancer.

Most naked mAbs attach to antigens on cancer cells, but some work by binding to antigens on non-cancerous cells or even free-floating proteins. Naked mAbs can work in different ways, such as:

- Boosting the immune response against cancer cells by attaching to them and acting as a marker for the body's immune system to destroy them. **Rituximab (Rituxan)**, used to treat some types of non-Hodgkin lymphoma (NHL), is an example.
- Improving the immune response by targeting [immune system checkpoints](#). **Pembrolizumab (Keytruda)**, used to treat some types of non-small cell lung cancer, is an example.
- Attaching to and blocking proteins on cancer cells (or other nearby cells) that help cancer cells grow or spread. **Trastuzumab (Herceptin)**, used in some types of breast cancer, is an example.

Conjugated monoclonal antibodies

Conjugated mAbs are connected to a chemotherapy drug or a radioactive particle. These mAbs act as a homing device to take one of these substances directly to the cancer cells. The mAb circulates throughout the body until it can find and hook onto the target antigen. It then delivers the toxic substance where it is needed most. Conjugated mAbs are also sometimes referred to as **tagged, labeled, or loaded antibodies**.

Antibody-drug conjugates (ADCs): These mAbs have powerful chemotherapy (or other) drugs attached to them. Examples include:

- **Brentuximab vedotin (Adcetris)**, an ADC made up of an antibody that targets the CD30 antigen (found on lymphocytes), attached to a chemo drug called *MMAE*
- **Ado-trastuzumab emtansine (Kadcyla, also called TDM-1)**, an ADC made up of an antibody that targets the HER2 protein, attached to a chemo drug called DM1

Radiolabeled antibodies: Radiolabeled antibodies have small radioactive particles attached to them. The antibody delivers radioactivity directly to cancer cells. Treatment with this type of antibody is sometimes known as [radioimmunotherapy \(RIT\)](#)². The drug and radiation are delivered directly to the target cells because the mAb looks for the target, and then the radiation affects the target and nearby cells to a certain extent.

Bispecific monoclonal antibodies

These drugs are made up of parts of 2 different mAbs, meaning they can attach to 2 different proteins at the same time.

T-cell engagers (TCEs): In these drugs, 1 part attaches to a protein on cancer cells, and the other sticks to a protein on immune cells called **T cells**. This brings the immune cells into contact with the cancer cells, which helps the immune system mount a more effective response against them.

Side effects of monoclonal antibodies

Monoclonal antibodies are most often given as an infusion into a vein (IV). The antibodies themselves are proteins, and giving them can sometimes cause an [infusion reaction](#)³, which is something like an allergic reaction. This is more common while the drug is first being given. Possible symptoms can include:

- [Fever](#)⁴
- Chills
- [Weakness](#)⁵
- [Headache](#)⁶
- [Nausea and vomiting](#)⁷
- [Diarrhea](#)⁸
- Low blood pressure
- [Skin changes](#)⁹ such as rashes

Compared with chemotherapy drugs, naked mAbs tend to have fewer serious side effects. But they can still cause problems in some people.

Some mAbs can have side effects that are related to the antigens they target. For example:

- Bevacizumab (Avastin) is an mAb that targets a protein called *VEGF* that affects tumor blood vessel growth. It can cause side effects such as high blood pressure, [bleeding](#)¹⁰, [poor wound healing](#)¹¹, [blood clots](#)¹², and kidney damage.
- Cetuximab (Erbix) is an antibody that targets a cell protein called *EGFR*, which is found on normal skin cells (as well as some types of cancer cells). This drug can cause serious rashes in some people.

Your cancer care team will watch you closely during treatment and will check you often. Side effects can and should be treated as early as possible. It's important that you tell your cancer care team about any changes in how you feel or anything you notice that's new or unusual. Tell them right away so they can treat any problems and try to keep them from getting worse.

Hyperlinks

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CAR T-cell Therapy and Its Side Effects

CAR T-cell therapy is a personalized form of immunotherapy that trains your own immune cells to recognize and destroy cancer. It can be a powerful option for treating certain hard-to-treat blood cancers, especially when other treatments are no longer effective.

[What is CAR T-cell therapy?](#)

[How does CAR T-cell therapy work?](#)

[Which cancers can be treated with CAR T-cell therapy?](#)

[What is it like to have CAR T-cell therapy?](#)

[Side effects of CAR T-cell therapy](#)

[How is CAR T-cell therapy paid for?](#)

[More resources from the American Cancer Society](#)

What is CAR T-cell therapy?

CAR T-cell therapy or **CAR T therapy** (short for **chimeric antigen receptor T-cell therapy**) is a type of cell-based gene therapy that alters the genes in a person's T cells (a type of white blood cell) to help them attack cancer cells.

This type of treatment can be helpful in treating some types of cancer, even when other treatments are no longer working.

How does CAR T-cell therapy work?

To understand how CAR T-cell therapy works, it helps to know how the immune system uses receptors on T cells to find antigens on cells it sees as foreign. This connection is key to how CAR T cells are made to target cancer.

How immune receptors and foreign antigens interact

The immune system recognizes foreign substances in the body by finding proteins called **antigens** on the surface of those cells. Immune cells called **T cells** (sometimes called **T lymphocytes**) have their own proteins called **receptors** that attach to foreign antigens and help trigger other parts of the immune system to destroy the foreign substance.

The relationship between antigens and immune receptors is like a lock and key. Just as a lock can only be opened with the right key, each foreign antigen has a unique immune receptor that can bind to it.

Cancer cells also have antigens, but if your immune cells don't have the right receptors, they can't attach to the antigens and help destroy the cancer cells.

How CAR T-cell therapy changes your T cells

In CAR T-cell therapy, T cells are taken from your blood and genetically changed in the lab to have a receptor called a **chimeric antigen receptor (CAR)** on their outer surface. This helps the T cells attach to a specific cancer cell antigen. The CAR T cells are then returned to your blood.

Since different cancers have different antigens, each CAR is made for a specific cancer's antigen. This means CAR T-cell therapies are **monoclonal**—each is designed to recognize one target.

For example, in certain kinds of leukemia or lymphoma, the cancer cells have an antigen called CD19. Many of the CAR T-cell therapies to treat these cancers are made to attach to the CD19 antigen and will not work for a cancer that doesn't have the CD19 antigen.

This is different from **polyclonal** cellular therapies such as [tumor infiltrating lymphocyte \(TIL\) therapy](#) that can recognize multiple antigens.

Which cancers can be treated with CAR T-cell therapy?

CAR T-cell therapies are approved by the US Food and Drug Administration (FDA) to treat some kinds of lymphomas and leukemias, as well as multiple myeloma. CAR T-cell therapy is typically used after other types of treatment have been tried.

Examples of CAR T-cell therapies currently approved include:

- **Tisagenlecleucel**, also known as **tisa-cel (Kymriah)**
- **Axicabtagene ciloleucel**, also known as **axi-cel (Yescarta)**
- **Brexucabtagene autoleucel**, also known as **brexu-cel (Tecartus)**
- **Lisocabtagene maraleucel**, also known as **liso-cel (Breyanzi)**
- **Idecabtagene vicleucel**, also known as **ide-cel (Abecma)**
- **Ciltacabtagene autoleucel**, also known as **cilta-cel (Carvykti)**
- **Obecabtagene autoleucel**, also known as **obe-cel (Aucatzyl)**

Many other CAR T-cell therapies (and similar types of treatment) are now being studied in clinical trials, in the hope of treating other types of cancer as well.

What is it like to have CAR T-cell therapy?

Getting CAR T-cell therapy is quite different from other types of immunotherapy treatment. It involves:

- Having the T cells collected from your blood, usually in 1 or more outpatient procedures
- Waiting several weeks for them to be modified and multiplied in the lab
- Getting the CAR T-cell infusion and being monitored closely, sometimes in the hospital for 1-2 weeks

Collecting the T cells

First, white blood cells (which include T cells) are removed from your blood using a procedure called **leukapheresis**. Two IV lines are used:

- Blood is removed through one line
- The white blood cells are separated out using a special machine (**apheresis**)
- Then the remaining blood is put back into your body through the other line.

Sometimes a special type of IV line called a [central venous catheter](#)¹ is used, which has both IV lines built in.

This process typically takes several hours, and you will need to stay seated or lying down during the procedure. It might need to be repeated another day or more until enough cells are collected.

Sometimes blood calcium levels can drop during leukapheresis, which can cause numbness and tingling or muscle spasms. This can be treated by replacing the calcium, which may be given by mouth or through an IV.

Waiting for the CAR T cells to be made

After the white cells are removed, they are frozen and sent to a special cell therapy lab. From there they are genetically changed to have the specific chimeric antigen receptor (CAR) on the surface of the T cell. This makes them **CAR T cells**.

These cells are then grown and multiplied in the lab. It can take several weeks to make the large number of CAR T cells needed for this therapy.

Receiving the CAR T-cell infusion

Once enough CAR T cells have been made, steps will be taken to get you ready to receive them.

A few days before the CAR T-cell infusion, you will likely be given a short course of [chemotherapy](#)² to help lower the number of other immune cells in your body. This gives the CAR T cells a better chance to get activated to destroy cancer without your body attacking them. This chemotherapy is usually not very strong because CAR T cells work best when there are still some cancer cells to attack.

A few days later, you will receive your CAR T cells as an infusion (through an IV) into your blood. This usually takes between 30 to 60 minutes, but can last longer. You'll be monitored for any signs of a reaction. Once the infusion is complete, your doctor will tell you when you can return home.

For several weeks after the infusion, you'll get regular checks at the hospital. This will help your doctor identify any problems or reactions to the therapy as soon as possible.

Once the CAR T cells start binding with cancer cells, they start to increase in number and can help destroy even more cancer cells.

Side effects of CAR T-cell therapy

CAR T-cell therapy can be very effective against some types of hard-to-treat cancers, but it can also sometimes cause serious or even life-threatening side effects. Because of this, it needs to be given in a medical center specially trained in its use, and you will need to be watched closely for several weeks after getting the CAR T cells.

Cytokine release syndrome (CRS)

As CAR T cells multiply, they can release large amounts of chemicals called **cytokines** into the blood, which can ramp up the immune system. Serious side effects from this release can include:

- High [fever](#)³ and chills
- Trouble breathing
- Severe [nausea, vomiting](#)⁴, and/or [diarrhea](#)⁵
- Feeling dizzy or lightheaded
- Headaches
- Fast heartbeat
- [Feeling very tired](#)⁶

As doctors are gaining more experience with CAR T-cell therapy, they are learning how to recognize CRS early, as well as how to treat it.

Nervous system problems

This treatment can sometimes have serious effects on the nervous system, leading to a condition called **immune effector cell-associated neurotoxicity syndrome (ICANS)**. This can result in symptoms such as:

- Headaches

- Changes in consciousness (awake easily or hard to awaken)
- Confusion or agitation
- Seizures
- Shaking or twitching (tremors)
- Trouble speaking and understanding
- Loss of balance

Because of the risk of these side effects, you'll likely be told not to drive, operate heavy machinery, or do any other potentially dangerous activities for at least 8 weeks after getting treatment.

Other serious side effects

Other possible serious side effects of CAR T-cell therapy can include:

- Allergic reactions during the infusion
- Abnormal levels of minerals in the blood, such as low potassium, sodium, or phosphorous levels
- A weakened immune system, with an increased risk of serious infections
- [Low blood cell counts](#)⁷, which can increase the risk of infections, fatigue, and bruising or bleeding
- Increased risk of [second cancers](#)⁸

If you are getting CAR T-cell therapy, it's very important to report any side effects to your cancer care team right away, as there are often medicines that can help treat them.

How is CAR T-cell therapy paid for?

CAR T therapy is expensive. On average, the cost of the CAR T cells alone can range from \$300,000 to \$475,000 because of the complexity of how the cells are collected and made. This doesn't include the costs of hospital admission, tests, procedures, and other expenses which can lead to the total cost being well over \$500,000.

While Medicare and many commercial insurances do cover some of the costs of CAR T therapy, they usually don't cover the full amount. The amount of coverage also varies between commercial plans. Medicaid coverage also varies and depends on your state of residence.

Before starting CAR T therapy, your care team will work with your insurance company to check what your policy covers. You will want to ask questions to learn as much as possible about the [costs of cancer treatment](#)⁹. Be sure to talk with them about out-of-pocket costs including deductibles, co-payments as well as other medication costs. The CAR T center will also likely have a social worker, nurse navigator or financial coordinator to help with this process. They can talk with you about [programs to help with expenses](#)¹⁰ and help you access resources in your local community.

More resources from the American Cancer Society

[Watch this video on YouTube](#)¹¹

[Flyer: CAR T-cell Therapy for Cancer](#)¹²

Learn how CAR T-cell therapy works to treat cancer and find a list of questions to ask.

[Flyer: Resources for CAR T-cell Therapy Patients & Caregivers](#)¹³

If you or someone you know is getting CAR T-cell therapy, the American Cancer Society offers information and support every step of the cancer journey.

Hyperlinks

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Last Revised: July 7, 2025

Tumor Infiltrating Lymphocyte Therapy and Its Side Effects

Tumor infiltrating lymphocyte (TILs) therapy or **tumor-derived autologous T cell immunotherapy** is a form of immunotherapy that uses your body's own immune cells to target and destroy cancer. It involves growing and reintroducing special immune cells taken directly from your tumor. TILs therapy can be very helpful in treating some types of solid tumor cancers, such as melanoma, even when other treatments are no longer working.

[What are tumor-infiltrating lymphocytes \(TILs\)?](#)

[What is TILs therapy and how does it work?](#)

[What is the difference between TILs therapy and CAR T-cell therapy?](#)

[Which cancers can be treated with TILs therapy?](#)

[Getting TILs therapy](#)

[Side effects of TILs therapy](#)

[How is TILs therapy paid for?](#)

What are tumor-infiltrating lymphocytes (TILs)?

Tumor-infiltrating lymphocytes (TILs) are a part of the immune system. They are specific types of white blood cells (**lymphocytes**) that have left the bloodstream and entered (infiltrated) a tumor to attack the cancer cells. Once TILs get inside the tumor, they can find different targets on the cancer cells and destroy them.

What is TILs therapy and how does it work?

TILs therapy helps your immune system fight cancer by using your own tumor-infiltrating lymphocytes (TILs). These cells are collected from your tumor, multiplied in a lab, and then returned to your body in large numbers to help target and destroy cancer cells.

To understand how TILs therapy can help treat cancer, it helps to know how the immune system works to find foreign or harmful substances like germs or cancer cells.

How immune receptors and foreign antigens interact

The immune system recognizes foreign substances by looking for **antigens** – proteins on the surface of those cells. Immune cells called **lymphocytes (T cells and B cells)** have their own proteins called **receptors** that attach to these antigens and help trigger other parts of the immune system to destroy the foreign substance.

The relationship between antigens and immune receptors is like a lock and key. Just as a lock can only be opened with the right key, each foreign antigen has a unique immune receptor that can bind to it.

Cancer cells also have antigens, but if your immune cells don't have the right receptors, they can't attach to the antigens and help destroy the cancer cells.

How TILs therapy boosts your immune response

For TILs therapy, a piece of your tumor is removed and analyzed in a cell therapy lab to find the lymphocytes (TILs) that have naturally found and entered the tumor, showing they carry the right receptors to recognize the antigens on your cancer cells. The TILs are carefully tested to choose the ones that respond most strongly to your cancer cells.

These are then multiplied in the lab and given back to you in large numbers to help your immune system kill the cancer cells.

What is the difference between TILs therapy and CAR T-cell therapy?

TILs therapy and **CAR (chimeric antigen receptor) T-cell therapy** are both cellular therapies that stimulate your immune system to find and attack cancer cells. However, there are differences:

- TILs are used to treat solid tumors, while CAR T cells are used to treat blood cancers like lymphoma, leukemia, and multiple myeloma.
- TILs are collected directly from the tumor, while CAR T cells are collected from the bloodstream.
- TILs can already recognize the cancer cells, so no special changes must be made to the cells once they are removed. CAR T cells must be modified in the lab to recognize and attack the cancer cells.
- TILs therapy is **polyclonal**, meaning it can attack many different antigens in the cancer cells, while CAR T cell therapy is monoclonal, meaning it targets one specific cancer antigen.

Which cancers can be treated with TILs therapy?

One type of TILs therapy called **Lifileucel (Amtagvi)** is approved by the US Food and Drug Administration (FDA) to treat [melanoma](#)¹ that has spread to other parts of the body. TILs therapy is typically used after other types of treatment have been tried.

Many other TILs therapies (and similar types of treatment) are now being studied in [clinical trials](#)², in the hope of treating other types of solid tumors such as lung, cervical, and genitourinary cancers.

Getting TILs therapy

Getting TILs therapy is quite different from other types of immunotherapy treatment. It involves:

- Surgery to remove a piece of the tumor
- Waiting several weeks for the TILs to be identified and multiplied in the lab
- Getting the TILs infusion and being monitored closely, sometimes in the hospital for 1-2 weeks

Collecting and growing the TILs

First, a part of your tumor is removed during surgery. It is sent to a special lab where the lymphocytes are separated out from the other cells. The TILs are identified and then stimulated to grow to large numbers over a period of several weeks.

Receiving the TILs infusion

Once enough TILs have grown, steps will be taken to get you ready to receive them.

A few days before the TILs infusion, you will likely be given [chemotherapy](#)³ to help lower the number of other immune cells in your body. This helps the body accept the TILs. This may be done in the hospital or in the clinic, depending on the center.

Following a few days of “rest,” you’ll be admitted to the hospital to receive the TILs infusion (through an IV) into your blood. This usually takes a few hours. After getting the TILs, you may also be given a type of immunotherapy called interleukin-2 (IL-2), which will help the TILs attack the cancer. This is usually done several hours after the TIL infusion and may be repeated over several days. You’ll be monitored closely for any signs of a reaction during this time and for several days afterwards. Sometimes you’ll need to stay in the hospital for a few weeks before going home.

For several weeks, you’ll need to get regular checks at the hospital or clinic. This will help your doctor identify any problems or reactions to the therapy as soon as possible.

Side effects of TILs therapy

TILs therapy can be very effective against some hard-to-treat cancers, but it can also sometimes cause serious or even life-threatening side effects. Because of this, it needs to be given in a medical center that is specially trained in its use, and you will need to be watched closely for several weeks after getting the TILs cells.

Serious side effects can include:

- Very [low blood cell counts](#)⁴, which can increase the risk of serious bleeding and infections
- Internal bleeding
- Serious infections
- Heart problems
- Lung problems
- Kidney problems
- [Infusion reactions](#)⁵

TILs therapy can also cause **capillary leak syndrome** that causes proteins and fluid to leak out of small blood vessels (capillaries) into the tissue. This can cause a decrease in blood pressure and if it worsens, can lead to organ failure.

If you are getting TILs therapy, it's very important to report any side effects to your health care team right away, as there are often medicines that can help treat them.

How is TILs therapy paid for?

TILs therapy is expensive. The cost of the TILs cells alone is around \$500,000 because of the complexity of how the TILs are collected and made. This doesn't include the costs of hospital admission, surgery needed to collect the specimen used to identify the TILs, tests, procedures, and other expenses which can lead to the total cost being well over \$500,000. While Medicare and some commercial insurances do cover some of the costs of TILs therapy, they usually don't cover the full amount. The amount of coverage also varies between commercial plans. Medicaid coverage also varies and depends on your state of residence.

Before starting TILs therapy, your care team will work with your insurance company to check what your policy covers. You will want to ask questions to learn as much as possible about the [costs of cancer treatment](#)⁶. Be sure to talk with them about out-of-pocket costs including deductibles, co-payments as well as other medication costs. The TILs center will also likely have a social worker, nurse navigator or financial coordinator to help with this process. They can talk with you about [programs to help with expenses](#)⁷ and help you access resources in your local community.

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2. www.cancer.org/cancer/managing-cancer/making-treatment-decisions/clinical-trials.html
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5. www.cancer.org/cancer/managing-cancer/side-effects/infusion-immune-reactions.html
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Immune Checkpoint Inhibitors and Their Side Effects

The immune system uses checkpoint proteins on immune cells to tell the difference between normal cells and those it sees as foreign, such as germs and cancer cells.

These checkpoints act like switches that turn immune responses on or off, helping the body avoid attacking its own tissues.

Some cancer cells find ways to use these checkpoints to avoid being recognized and destroyed. **Immune checkpoint inhibitors** block the signals from these checkpoints, allowing the immune system to find and attack the cancer cells.

[What are immune checkpoint inhibitors?](#)

[PD-1 and PD-L1 inhibitors](#)

[CTLA-4 inhibitors](#)

[LAG-3 inhibitors](#)

[Side effects of checkpoint inhibitors](#)

What are immune checkpoint inhibitors?

An **immune checkpoint inhibitor** (or just **checkpoint inhibitor**) is a type of **monoclonal antibody**, a lab-made protein designed to block checkpoint proteins that act like switches to turn the immune response on or off.

Checkpoint inhibitors don't kill cancer cells directly. They work by helping the immune system to better find and attack the cancer cells, wherever they are in the body. Medicines that target different checkpoint proteins are now used to treat many types of cancer.

Most of these drugs are given as an infusion into a vein (IV). Some of them can be given as an injection under the skin (subcutaneously) over several minutes as well. Your doctor will recommend which way of giving the checkpoint inhibitor is best for you.

PD-1 and PD-L1 inhibitors

PD-1 is a checkpoint protein on immune cells called **T cells**. It acts as a “off switch” to help keep the T cells from attacking normal cells. PD-1 does this by attaching to PD-

L1, a protein found on some normal and cancer cells. When PD-1 binds to PD-L1, it tells the T cell not to attack. Some cancer cells have large amounts of PD-L1, which helps them avoid being attacked by the immune system.

Monoclonal antibodies that target either PD-1 or PD-L1 can block this binding and boost the immune response against cancer cells.

Drugs that target PD-1 include:

- **Pembrolizumab (Keytruda and Keytruda Qlex)**
- **Nivolumab (Opdivo and Opdivo Qvantig)**
- **Cemiplimab (Libtayo)**

Drugs that target PD-L1 include:

- **Atezolizumab (Tecentriq and Tecentriq Hybreza)**
- **Avelumab (Bavencio)**
- **Durvalumab (Imfinzi)**

Both PD-1 and PD-L1 inhibitors have been shown to be helpful in treating many different types of cancer.

CTLA-4 inhibitors

CTLA-4 is another checkpoint protein on some T cells that acts as an “off switch” to help keep the immune system in check.

Ipilimumab (Yervoy) and **tremelimumab (Imjudo)** are monoclonal antibodies that attach to CTLA-4 and stop it from working. This can help boost the body’s immune response against cancer cells.

These drugs are typically used along with a PD-1 or PD-L1 inhibitor to treat some types of cancer.

LAG-3 inhibitors

LAG-3 is a checkpoint protein on some types of immune cells that normally acts as an “off switch” to help keep the immune system in check.

Relatlimab is a monoclonal antibody that attaches to LAG-3 and stops it from working. This can help boost the body’s immune response against cancer cells.

This drug is given along with the PD-1 inhibitor nivolumab (in a combination known as **Opdualag**). It can be used to treat [melanoma of the skin](#)¹, and it’s being studied for use in several other types of cancer.

Side effects of checkpoint inhibitors

Side effects of checkpoint inhibitors vary from person to person and drug to drug. You may hear these referred to as **immune-related adverse events** by your care team. They also can range from mild to severe, and quick recognition is important. Some of the more common side effects of checkpoint inhibitors include:

- [Diarrhea](#)²
- [Fatigue](#)³
- Cough
- [Nausea and vomiting](#)⁴
- [Skin rash](#)⁵
- [Poor appetite](#)⁶
- [Constipation](#)⁷
- [Muscle and joint pain](#)⁸

Other, more serious side effects occur less often:

[Infusion reactions](#)⁹: Some people might have an infusion reaction while getting these drugs. This is like an allergic reaction, and can include fever, chills, flushing of the face, rash, itchy skin, feeling dizzy, wheezing, and trouble breathing. It’s important to tell your doctor or nurse right away if you have any of these symptoms while getting one of these drugs.

Autoimmune reactions: By targeting a checkpoint protein, these drugs remove one of the safeguards on the body's immune system. Sometimes the immune system responds by attacking other parts of the body, which can cause serious or even life-threatening problems in the lungs, intestines, liver, hormone-making glands, kidneys, or other organs.

Cytokine release syndrome (CRS): This is a rare side effect of checkpoint inhibitors. They might cause immune cells to release large amounts of chemicals called **cytokines** into the blood, which can further ramp up the immune system. Serious side effects from this release can include:

- High [fever](#)¹⁰ and chills
- Trouble breathing
- Severe nausea, vomiting, and/or diarrhea
- Feeling dizzy or lightheaded
- Headaches
- Fast heartbeat
- Feeling very tired

It's very important to report any new side effects to someone on your health care team as soon as possible. If serious side effects do occur, treatment may need to be stopped and you might be given high doses of corticosteroids to suppress your immune system.

Your cancer care team will watch you closely during treatment and will check you often. Side effects can and should be treated as early as possible. It's important to tell your cancer care team about any changes in how you feel or anything you notice that's new or unusual. Tell them right away so they can treat any problems and try to keep them from getting worse.

Hyperlinks

1. www.cancer.org/cancer/types/melanoma-skin-cancer.html
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Cancer Vaccines and Their Side Effects

Most people are familiar with vaccines used to prevent infections like measles and chickenpox. They work by training the immune system to recognize and respond to harmful germs. But there are also some vaccines that can also be designed to help the immune system find and attack cancer cells. Cancer vaccines are currently available for some prostate and bladder cancers, and more are being researched.

[What are cancer vaccines?](#)

[How are vaccines used to treat cancer?](#)

[What are cancer vaccines made of?](#)

[Types of therapeutic cancer vaccines](#)

[Side effects of cancer vaccines](#)

[Why don't cancer vaccines work for everyone?](#)

[Ongoing therapeutic cancer vaccine research](#)

What are cancer vaccines?

Cancer vaccines are substances made in the lab that are used to make the body's natural defense mechanisms stronger to protect itself. Depending on the type of vaccine, some can be used as cancer treatment and other times, as cancer prevention (such as the [human papillomavirus, or HPV, vaccine¹](#)).

How are vaccines used to treat cancer?

Vaccines used to treat cancer are called **therapeutic cancer vaccines**. Instead of preventing disease, they are meant to get the immune system to attack a disease a person already has.

Therapeutic cancer vaccines can work by:

- Keeping cancer from coming back
- Destroying any cancer cells still in the body after treatments end
- Stopping a tumor from growing or spreading

Therapeutic cancer vaccines teach the immune system to find and attack cancer cells that have certain markers, called **antigens**. The immune system can remember these markers, so the vaccine might keep working long after it's given.

What are cancer vaccines made of?

Some cancer vaccines are made from cancer cells, parts of cells, or pure antigens (certain proteins on the cancer cells). Sometimes a person's own immune cells are removed and exposed to these substances in the lab to create the vaccine.

Vaccines are often combined with other substances or cells called **adjuvants** that help boost the immune response even further.

Other vaccines are made from bacteria or viruses that can trigger a specific immune response.

Types of therapeutic cancer vaccines

Several therapeutic cancer vaccines have been approved by the US Food and Drug Administration (FDA). They include:

Sipuleucel-T (Provenge) is used to treat advanced prostate cancer that is no longer being helped by hormone therapy. This vaccine is made by:

- Removing immune cells (white blood cells) from your blood
- Combining them with a protein from prostate cancer cells
- Giving them back to you by infusing them into a vein (IV)

The process is repeated a total of 3 times. The cells then work to help your immune system attack the prostate cancer cells. While it hasn't been shown to cure cancer, it can help some people live longer.

Bacillus Calmette-Guerin (BCG) is used to treat early-stage bladder cancer. It's made from a type of bacteria related to the one that causes tuberculosis, but it doesn't cause someone to get sick. It is given through an [intravesical infusion](#)² directly into the bladder, where it stimulates the immune system to attack the bladder cancer cells.

Nadofaragene firadenovec (Adstiladrin) is a gene therapy that works much like a cancer vaccine to treat early-stage bladder cancers that have progressed after BCG therapy. It uses a virus to deliver a gene that stimulates bladder cells to produce interferon alfa-2b, a protein that helps the immune system attack cancer cells. Like BCG, it's given through an intravesical infusion directly into the bladder.

Side effects of cancer vaccines

Side effects of cancer vaccines differ from person to person and depend on the type of vaccine that is given. Some common side effects include:

- [Fever](#)³
- Chills
- [Fatigue](#)⁴
- Back and joint pain
- [Nausea](#)⁵
- [Headache](#)⁶
- Dizziness

[Infusion reactions](#)⁷ most often start suddenly, usually within minutes to hours after taking the immunotherapy. They may include hives (raised skin welts that often go away in a day or so) and intense itching. An allergic or infusion reaction often includes other serious symptoms such as trouble breathing, dizziness, tightness in the throat or chest, or swelling of the lips or tongue.

Other, more serious side effects occur less often and include:

- Difficulty breathing
- High blood pressure

- Stroke

Your cancer care team will watch you closely during treatment and will check you often. Side effects can and should be treated as early as possible. It's important to tell your cancer care team about any changes in how you feel or anything you notice that's new or unusual. Tell them right away so they can treat any problems and try to keep them from getting worse.

Why don't cancer vaccines work for everyone?

Making therapeutic cancer vaccines that work is a challenge because:

Cancer cells suppress the immune system. This is how cancer begins and grows in the first place. Researchers are using adjuvants in vaccines to try to fix this problem.

Cancer cells start from a person's own healthy cells. As a result, cancer cells may not look harmful to the immune system. The immune system may ignore the cells instead of finding and attacking them.

Larger or more advanced tumors are hard to get rid of using only a vaccine. This is one reason doctors often give a cancer vaccine along with other treatment.

People who are sick or older can have weak immune systems. Their bodies may not be able to produce a strong immune response after they receive a vaccine. That limits how well a vaccine works. Also, some cancer treatments may weaken a person's immune system. This limits how well the body can respond to a vaccine.

For these reasons, some researchers think cancer treatment vaccines may work better for smaller tumors or cancer in its early stages.

Ongoing therapeutic cancer vaccine research

Cancer vaccine research is quickly changing. Researchers are finding improved ways to identify targets for new therapeutic cancer vaccines that can work better to destroy cancer cells. This includes work in mRNA technology. They also are looking for ways to combine therapeutic cancer vaccines with other forms of cancer treatments, including another type of immunotherapy called checkpoint inhibitor therapy, to help the immune system work even better.

Learn more about [mRNA vaccines](#) used in cancer treatment.

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mRNA Vaccines

Scientists are studying whether vaccines made of mRNA may be useful in cancer treatment. Learn about mRNA vaccines, how they may help treat cancers, and what we know about their safety.

[Understanding mRNA](#)
[Why mRNA matters in cancer](#)
[Are mRNA vaccines safe?](#)
[Key points](#)

Understanding mRNA

mRNA stands for “messenger RNA.” It is a natural part of our cells that carries a temporary set of instructions from part of our DNA (our genes). It tells the cell how to make a certain protein, and then it is broken down.

What mRNA does:

- Gives short-term instructions for the body to make a protein
- Works only in the outer part of the cell (not in the nucleus of the cell, where DNA lives)
- Breaks down within hours to days

What mRNA does NOT do:

- Does not affect your DNA, where your genes are
- Does not stay in your body long-term
- Cannot copy or spread on its own

Why mRNA matters in cancer

Using mRNA is a promising new way to make [vaccines to help treat cancer](#).

Vaccines are substances that help your immune system learn to fight off something that doesn't belong there. We're most familiar with vaccines that help prevent some types of infections, such as the flu vaccine. But vaccines might also be helpful in treating cancer, if the vaccine can get the immune system to see and attack cancer cells. It's been hard to make effective cancer vaccines for several reasons, but mRNA might provide a way to make cancer vaccines that are more effective and easier to create.

mRNA vaccines for cancer treatment

When used as a vaccine, mRNA can be injected into the body packaged in tiny fat droplets. It finds its way into cells and gives them temporary instructions to make one or more proteins found on cancer cells. The immune system then learns to recognize those proteins as "flags" and attack the cancer cells that carry them.

mRNA vaccines might prove to be useful in several ways to treat cancer:

- **Personalized vaccines:** mRNA vaccines can be designed and made quickly, which is important because cancer cells can change frequently. Scientists may be able to design personalized mRNA vaccines that target unique properties of cancer

cells, called “neoantigens,” before the cancer changes. The immune system can attack the cancer cells with neoantigens without harming normal cells. Each cancer has a unique mix of neoantigens, so a personalized approach may be useful.

- **General vaccines:** Some mRNA vaccines are made to target cancer proteins that many patients share, so they could help a wider group of people.
- **Boosting other treatments:** mRNA vaccines might be helpful when combined with other types of immunotherapy, such as [checkpoint inhibitors](#).

Clinical trials are testing mRNA cancer vaccines in several types of cancer, including lung, breast, prostate, melanoma (skin), pancreatic, and brain cancers. Early results have been promising, but more research is still needed.

Are mRNA vaccines safe?

Each mRNA vaccine (or any other type of vaccine) has its own potential side effects.

So far, the only widely available mRNA vaccines have been used to help prevent [COVID-19](#)¹. The mRNA vaccines have been given to many people worldwide, and they've generally shown a good safety record. This has given doctors and scientists a lot of information about safety.

What we've seen so far:

- The most common reactions are short-term, such as soreness at the injection site, fatigue, or mild fever.
- Uncommon side effects, like heart inflammation in young men after COVID-19 vaccines, have been reported but remain rare.
- In cancer studies, the most common side effects are mild flu-like symptoms caused by the immune system becoming active.

How safety is checked:

- The manufacturing process is carefully controlled for purity and safety.
- New mRNA treatments need to go through several phases of [clinical trials](#)² before approval.
- After approval, safety continues to be monitored worldwide.

Key points

- mRNA vaccines are a promising area of cancer treatment research.
- Although none have been approved at this time, they're being studied for use against several types of cancer.
- mRNA vaccines can be made fairly quickly, which might allow for more personalized (and hopefully more effective) vaccines in the future.
- mRNA vaccines cannot change your DNA.
- mRNA vaccines cannot give you an infection.
- mRNA from a vaccine stays in the body for only a short time before it naturally breaks down.

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Cytokines and Their Side Effects

The immune system uses complex pathways to coordinate its response and protect the body from outside invaders. These include proteins and other molecules that help direct the immune response. Some of these can be used treatments to help the body kill cancer cells.

[What are cytokines?](#)[How do cytokines work?](#)[Types of cytokines used in cancer treatment](#)[Side effects of cytokines](#)

What are cytokines?

Cytokines are small proteins that help control the growth and activity of immune cells and blood cells. They are made by different types of white blood cells and are sometimes classified based which cells produce them:

- **Lymphokines** are made by lymphocytes
- **Monokines** are made by monocytes

How do cytokines work?

When released, cytokines signal the immune system to do its job. Cytokines affect the growth of all blood cells and other cells that help the body's immune and inflammation responses. They also help to boost anti-cancer activity by sending signals that can help make abnormal cells die and normal cells live longer.

Cytokines have multiple functions, based on their type:

- **Interleukins** act as chemical signals between white blood cells
- **Interferons** help the body resist virus infections and cancers
- **Tumor necrosis factors** signal immune cells to attack and kill cancer cells
- **Growth factors** support cell development and tissue repair
- **Chemokines** guide immune cells to where they're needed

Types of cytokines used in cancer treatment

Some cytokines can be made in a lab and are used to treat cancer. Others help prevent or manage chemotherapy side effects. Cytokines are injected, either under the skin, into a muscle, or into a vein.

The most common cytokines used in cancer care are interleukins and interferons. Depending on the type and stage of cancer, they can be used by themselves or with other treatments such as chemotherapy or other immunotherapy.

Interleukin-2 (IL-2)

Interleukins act as chemical signals between white blood cells. **Interleukin-2 (IL-2)** helps immune system cells grow and divide more quickly. A lab-made version of IL-2 called **proleukin (aldesleukin)** is approved to treat advanced [kidney cancer](#)¹ and metastatic [melanoma](#)². IL-2 can be used as a single drug treatment for these cancers, or it can be combined with chemotherapy or with other cytokines such as interferon-alfa.

If IL-2 is given in high doses, it must be done in a hospital due to the risk of serious side effects.

Other interleukins, such as IL-7, IL-12, and IL-21, continue to be studied as stand-alone cancer treatments and as adjuvants, combined with other treatments.

Interferon-alpha (IFN-alfa)

Interferons help the body resist virus infections and cancers. The types of interferon (IFN) are named after the first 3 letters of the Greek alphabet:

- IFN-alfa
- IFN-beta
- IFN-gamma

Only IFN-alfa is used to treat cancer. It boosts the ability of certain immune cells to attack cancer cells. It may also slow the growth of cancer cells directly, as well as the blood vessels that tumors need to grow.

IFN-alfa can be used to treat these cancers:

- Hairy cell leukemia
- Chronic myelogenous leukemia (CML)
- Follicular non-Hodgkin lymphoma
- Cutaneous (skin) T-cell lymphoma
- Kidney cancer
- Melanoma
- Kaposi sarcoma

Growth factors

Growth factors are most commonly used to manage the side effects of cancer treatment. Certain types called **colony-stimulating factors** work by signaling immature white blood cells to grow and develop into mature cells. This can help the body better fight off active infection.

Tumor necrosis factor (TNF)

Tumor necrosis factor (TNF) signals immune cells to attack and kill cancer cells. It is also involved with inflammatory processes within the body. It's sometimes used to treat rheumatoid arthritis and is actively being studied as a treatment for some cancer types.

Side effects of cytokines

Side effects of cytokines differ from person to person and depend on the type of cytokine used. Some common side effects include:

- Chills
- [Fever](#)³
- [Fatigue](#)⁴
- [Confusion](#)⁵
- [Nausea and vomiting](#)⁶
- [Diarrhea](#)⁷
- [Neutropenia](#)⁸ (low white blood counts)
- Increased risks of certain [infections](#)⁹
- [Hair loss](#)¹⁰
- [Skin rashes](#)¹¹

Less common and more serious side effects include:

- Low blood pressure
- Abnormal heartbeat
- Heart problems
- Chest pain

Side effects with interferon can be severe and can make treatment hard for many people to tolerate. Most side effects don't last long after the treatment stops, but fatigue can last longer. Other rare long-term effects include damage to nerves, including those in the brain and spinal cord.

Your cancer care team will watch you closely during treatment and will check you often. Side effects can and should be treated as early as possible. It's important to tell your cancer care team about any changes in how you feel or anything you notice that's new or unusual. Tell them right away so they can treat any problems and try to keep them from getting worse.

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Oncolytic Virus Therapy and Its Side Effects

Some viruses can be specially changed to infect and kill cancer cells while leaving healthy cells alone. These viruses can also activate the immune system, helping it recognize and attack other cancer cells. Oncolytic virus therapy is currently used to treat melanoma.

[What is oncolytic virus therapy?](#)

[What types of oncolytic virus therapy are available to treat cancer?](#)

[Side effects of oncolytic viruses](#)

What is oncolytic virus therapy?

Oncolytic virus therapy, sometimes just called virus therapy, uses viruses that have been changed in a lab to destroy cancer cells. A genetically modified version of the virus is injected into the tumor. When the viruses enter cancer cells, they make copies of themselves. As a result, the cancer cells burst and die.

As the cancer cells die, they release proteins that trigger your immune system to target any other cancer cells in your body with the same proteins as the dead cancer cells. The virus doesn't enter healthy cells.

What types of oncolytic virus therapy are available to treat cancer?

Currently, one type of oncolytic virus therapy is approved in the United States to treat cancer:

Talimogene laherparepvec (Imlygic) or T-VEC. This oncolytic virus therapy is approved to treat advanced melanoma that cannot be treated with surgery. It's used most often for people who cannot or choose not to receive any other recommended treatments. T-VEC is a modified version of the herpes simplex virus, which causes cold sores. It is injected directly into one or more melanoma tumors. (This method is called **intralesional therapy**.) This treatment can sometimes shrink these tumors, and it might also shrink tumors in other parts of the body.

Side effects of oncolytic viruses

Side effects of oncolytic viruses differ from person to person and depend on the type of oncolytic virus used. Some common side effects include:

- [Fever¹](#)
- Chills
- Muscle or joint aches
- Pain at the injection site
- [Fatigue²](#)

Your cancer care team will watch you closely during treatment and will check you often. Side effects can and should be treated as early as possible. It's important to tell your cancer care team about any changes in how you feel or anything you notice that's new or unusual. Tell them right away so they can treat any problems and try to keep them from getting worse.

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Immunomodulators and Their Side Effects

While other types of cancer immunotherapy act on specific parts of the immune system, **immunomodulators** work in broader, less targeted ways to boost the body's immune response to cancer or certain other diseases.

[What are immunomodulators?](#)

[How do immunomodulators work in treating cancer?](#)

[How are immunomodulators given?](#)

[Types of immunomodulators used as immunotherapy](#)

Side effects of immunomodulators

What are immunomodulators?

Immunomodulator drugs (iMiDs), sometimes called **biological response modifiers**, are a group of drugs that target the action of certain proteins as part of the immune system's response to kill cancer cells.

They are a broad group of medications that treat a variety of different conditions, including:

- Asthma
- Autoimmune disease
- Cancer
- Eczema
- Inflammatory bowel disease (Crohn's disease, ulcerative colitis)
- Infection

How do immunomodulators work in treating cancer?

Immunomodulators work to either “turn up” or “turn down” certain proteins in the body. When proteins are turned up, this can lead to an increased response of the immune system. In cancer treatment, this can be helpful in allowing the immune system to work more effectively to find and destroy cancer cells.

When proteins are turned down, this can lead to a decreased response of the immune system. This can be helpful in cancer treatment by:

- Helping the body overcome the defense mechanisms cancer cells use to try and protect themselves against the immune system
- Blocking the ways cancer cells grow and divide, limiting their ability to make more cancer cells
- Reducing the immune system's response to donated tissue or blood following a stem cell or bone marrow transplant

Other types of immunotherapy also have immunomodulatory action:

- **Immune checkpoint inhibitors** block checkpoint proteins that act as “off switches” in the body. This helps the immune system destroy the cancer cells
- **Cytokines** turn on immune cells to be more effective against cancer
- **CAR T-cell therapy** turns on white blood cells called T-cells so they can better kill cancer cells

How are immunomodulators given?

Depending on the type, iMiDs given for cancer are given as an infusion into a vein (IV), pills or capsules that you swallow (oral), subcutaneous injections under the skin (SQ), or cream that you apply to your skin (topical). Some can be given alone but most are used in combination with other cancer treatments, including chemotherapy or immunotherapy.

Types of immunomodulators used as immunotherapy

Thalidomide, lenalidomide, and pomalidomide

Thalidomide (Thalomid), lenalidomide (Revlimid), and pomalidomide (Pomalyst) are iMiDs used to treat multiple myeloma, sometimes in combination with corticosteroids. They are taken daily as pills, with breaks from treatment on certain days each month. Because these drugs can increase the risk of serious blood clots, they are often given along with aspirin or a blood thinner.

These drugs might cause severe birth defects when taken during pregnancy. Because of this, they have special requirements. You will need to:

- Enroll in a special program run by the drug company that makes them.
- Take steps to not become pregnant while undergoing treatment.
- Follow special handling precautions to keep them away from others.

Your doctor will talk with you more about the specific safety requirements.

Imiquimod

Imiquimod is a cream used to treat some very early-stage skin cancers (or precancers), especially if they are in sensitive areas such as on the face. It stimulates a local immune response against skin cancer cells.

The cream is applied to the skin anywhere from once a day to twice a week for several months. Some people have serious skin reactions to this drug.

Side effects of immunomodulators

Side effects of iMiDs differ from person to person and depend on the type of iMiD used.

Common side effects of thalidomide, lenalidomide, and pomalidomide include:

- [Diarrhea](#)¹ or [constipation](#)²
- [Fatigue](#)³
- [Skin rashes](#)⁴
- [Low blood counts](#)⁵
- Increased risks of certain [infections](#)⁶
- [Neuropathy](#)⁷
- Increased risk of [blood clots](#)⁸

Common side effects of imiquimod include:

- Redness, burning, or irritation at the treatment site
- [Dryness or flakiness of the skin](#)⁹
- [Pain](#)¹⁰
- [Swelling](#)¹¹

Your cancer care team will watch you closely during treatment and will check you often. Side effects can and should be treated as early as possible. It's important to tell your cancer care team about any changes in how you feel or anything you notice that's new or unusual. Tell them right away so they can treat any problems and try to keep them from getting worse.

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Written by

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